

Multivariate Linear Regression

Readings: Johnson & Wichern 2007, Chapter 7; DSA 8020
Lectures 1–4 [Link](#); Zelterman, 2015, Chapter 9

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Agenda

Motivation

Model and Assumptions

Parameter Estimation

Inference and Prediction

Motor Trend Car Road Tests Data Analysis

Example: Motor Trend Car Road Tests

```
> head(mtcars)
```

	mpg	cyl	disp	hp	drat	wt	qsec	vs	am	gear	carb
Mazda RX4	21.0	6	160	110	3.90	2.620	16.46	0	1	4	4
Mazda RX4 Wag	21.0	6	160	110	3.90	2.875	17.02	0	1	4	4
Datsun 710	22.8	4	108	93	3.85	2.320	18.61	1	1	4	1
Hornet 4 Drive	21.4	6	258	110	3.08	3.215	19.44	1	0	3	1
Hornet Sportabout	18.7	8	360	175	3.15	3.440	17.02	0	0	3	2
Valiant	18.1	6	225	105	2.76	3.460	20.22	1	0	3	1

Suppose we would like to study the (linear) relationship between mpg, disp, hp, wt (**responses**) and cyl, am, carb (**predictors**)

Multiple regression predicts one outcome; **multivariate regression** predicts several simultaneously

Why Multivariate Regression?

Why not simply fit d separate regressions?

Estimation

Same OLS coefficient estimates



Joint Inference

Accounts for correlated responses

$$\hat{\beta}_k = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{y}_k$$

$$H_0 : \beta_{j1} = \cdots = \beta_{jd} = 0$$

Key idea

Separate regressions give the same **point estimates**.
But multivariate regression models the **dependence among responses**.

$\Rightarrow$ This matters for joint inference and prediction.

Review: Linear Regression Model

The multiple linear regression model has the form:

$$y_i = \beta_0 + \sum_{j=1}^p \beta_j x_{ij} + \varepsilon_i, \quad i = 1, \dots, n,$$

where

- ▶ y_i is the **response** for the i -th observation
- ▶ x_{ij} is the j -th **predictor** for the i -th observation
- ▶ β_0 and β_j 's are the **regression intercept** and **slopes** for the response, respectively
- ▶ ε_i is the **error** term for the response of the i -th observation

From One Response to Several Responses

Multiple Regression

One response

$$y_i = \beta_0 + \beta_1 x_{i1} + \cdots + \beta_p x_{ip} + \varepsilon_i$$

Multivariate Regression

Several responses

$$\mathbf{y}_i = \mathbf{B}^T \mathbf{x}_i + \boldsymbol{\varepsilon}_i$$

Predictors $\longrightarrow$ $\boxed{Y}$

Predictors $\longrightarrow$ $\boxed{Y_1, \dots, Y_d}$

The predictors are shared; each response has its own coefficients, while the response errors may be **correlated**.

The Multivariate Linear Regression Model: Scalar Form

The multivariate (multiple) linear regression model has the form:

$$y_{ik} = \beta_{0k} + \sum_{j=1}^p \beta_{jk} x_{ij} + \varepsilon_{ik}, \quad i = 1, \dots, n, \quad k = 1, \dots, d,$$

where

- ▶ y_{ik} is the k -th **response** for the i -th observation
- ▶ x_{ij} is the j -th **predictor** for the i -th observation
- ▶ β_{0k} and β_{jk} 's are the **regression intercept** and **slopes** for k -th response, respectively
- ▶ ε_{ik} is the **error** term for the k -th response of the i -th observation

The Multivariate Linear Regression Model: Assumptions

The assumptions of the model are:

- ▶ Relationship between $\{x_j\}_{j=1}^p$ and Y_k is **linear** for each $k \in \{1, \dots, d\}$
- ▶ $(\varepsilon_{i1}, \dots, \varepsilon_{id})^T \stackrel{i.i.d.}{\sim} N(\mathbf{0}, \Sigma)$ is an **unobserved random vector**
- ▶ $[Y_{ik} | x_{i1}, \dots, x_{ip}] \sim N(\beta_{0k} + \sum_{j=1}^p \beta_{jk} x_{ij}, \sigma_{kk})$ for each $k \in \{1, \dots, d\}$

What Is New in the Multivariate Model?

For observation i ,

$$\boldsymbol{\varepsilon}_i = \begin{bmatrix} \varepsilon_{i1} \\ \vdots \\ \varepsilon_{id} \end{bmatrix} \sim N_d(\mathbf{0}, \boldsymbol{\Sigma}).$$

$$\boldsymbol{\Sigma} = \begin{bmatrix} \sigma_{11} & \cdots & \sigma_{1d} \\ \vdots & \ddots & \vdots \\ \sigma_{d1} & \cdots & \sigma_{dd} \end{bmatrix}$$

- ▶ Diagonal: residual variation of each response
- ▶ Off-diagonal: residual dependence **between responses**
- ▶ Different observations i are assumed independent

The major extension from ordinary regression is modeling several responses jointly through $\boldsymbol{\Sigma}$.

Multivariate Regression: Matrix Form

The multivariate multiple linear regression model has the form

$$\mathbf{Y} = \mathbf{X}\mathbf{B} + \mathbf{E},$$

where

- ▶ $\mathbf{Y} = [\mathbf{y}_1, \dots, \mathbf{y}_d]$ is the $n \times d$ **response matrix**, where $\mathbf{y}_k = (y_{1k}, \dots, y_{nk})^T$ is the k -th response vector
- ▶ $\mathbf{X} = [\mathbf{1}, \mathbf{x}_1, \dots, \mathbf{x}_p]$ is the $n \times (p+1)$ **design matrix**
- ▶ $\mathbf{B} = [\boldsymbol{\beta}_1, \dots, \boldsymbol{\beta}_d]$ is the $(p+1) \times d$ **matrix of regression coefficients**
- ▶ $\mathbf{E} = [\boldsymbol{\varepsilon}_1, \dots, \boldsymbol{\varepsilon}_d]$ is the $n \times d$ **error matrix**

$$\underbrace{\mathbf{Y}}_{n \times d} = \underbrace{\mathbf{X}}_{n \times (p+1)} \underbrace{\mathbf{B}}_{(p+1) \times d} + \underbrace{\mathbf{E}}_{n \times d}$$

n : observations

p : predictors

d : responses

Another Look of the Matrix Form

Matrix form writes the multivariate linear regression model for all $n \times d$ points simultaneously as

$$\mathbf{Y} = \mathbf{X}\mathbf{B} + \mathbf{E}$$

$$\begin{bmatrix} y_{11} & \cdots & y_{1d} \\ y_{21} & \cdots & y_{2d} \\ \vdots & \ddots & \vdots \\ y_{n1} & \cdots & y_{nd} \end{bmatrix} = \begin{bmatrix} 1 & \cdots & x_{1p} \\ 1 & \cdots & x_{2p} \\ \vdots & \ddots & \vdots \\ 1 & \cdots & x_{np} \end{bmatrix} \begin{bmatrix} \beta_{01} & \cdots & \beta_{0d} \\ \beta_{11} & \cdots & \beta_{1d} \\ \vdots & \ddots & \vdots \\ \beta_{p1} & \cdots & \beta_{pd} \end{bmatrix} + \begin{bmatrix} \varepsilon_{11} & \cdots & \varepsilon_{1d} \\ \varepsilon_{21} & \cdots & \varepsilon_{2d} \\ \vdots & \ddots & \vdots \\ \varepsilon_{n1} & \cdots & \varepsilon_{nd} \end{bmatrix}$$

Assuming that n subjects are **independent**, we have

$$\blacktriangleright \varepsilon_k \sim \mathcal{N}(0, \sigma_{kk}), \quad k \in \{1, \dots, d\}$$

$$\blacktriangleright \varepsilon_i \stackrel{i.i.d.}{\sim} \mathcal{N}(\mathbf{0}, \Sigma), \quad i = 1, \dots, n$$

Ordinary Least Squares

The **ordinary least squares** OLS estimate is

$$\arg \min_{\mathbf{B} \in \mathbb{R}^{(p+1) \times d}} \|\mathbf{Y} - \mathbf{X}\mathbf{B}\|^2 = \arg \min_{\mathbf{B} \in \mathbb{R}^{(p+1) \times d}} \sum_{i=1}^n \sum_{k=1}^d \left(y_{ik} - \beta_{0k} - \sum_{j=1}^p \beta_{jk} x_{ij} \right)^2,$$

where $\|\cdot\|$ denotes the Frobenius norm.

$$\blacktriangleright \text{OLS}(\mathbf{B}) = \|\mathbf{Y} - \mathbf{X}\mathbf{B}\|^2 = \text{tr}(\mathbf{Y}^T \mathbf{Y}) - 2\text{tr}(\mathbf{Y}^T \mathbf{X}\mathbf{B}) + \text{tr}(\mathbf{B}^T \mathbf{X}^T \mathbf{X}\mathbf{B})$$

$$\blacktriangleright \frac{\partial \text{OLS}(\mathbf{B})}{\partial \mathbf{B}} = -2\mathbf{X}^T \mathbf{Y} + 2\mathbf{X}^T \mathbf{X}\mathbf{B}$$

The OLS estimate has the form

$$\hat{\mathbf{B}} = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{Y} \Rightarrow \hat{\beta}_k = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{y}_k, \quad k \in \{1, \dots, d\}$$

Expected Value of Least Squares Coefficients

The expected value of the estimated coefficients is given by

$$\begin{aligned}\mathbb{E}(\hat{\mathbf{B}}) &= \mathbb{E} \left[(\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{Y} \right] \\ &= (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbb{E}(\mathbf{Y}) \\ &= (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{X} \mathbf{B} \\ &= \mathbf{B}\end{aligned}$$

$\Rightarrow \hat{\mathbf{B}}$ is an unbiased estimator of $\mathbf{B}$

Fitted Values and Residuals

- Fitted values are given by

$$\hat{\mathbf{Y}} = \mathbf{X}\hat{\mathbf{B}},$$

$$\text{i.e., } \hat{y}_{ik} = \hat{\beta}_{0k} + \sum_{j=1}^p \hat{\beta}_{jk}x_{ij}, \quad i = 1, \dots, n, \quad k = 1, \dots, d$$

- Residuals are given by

$$\hat{\mathbf{E}} = \mathbf{Y} - \hat{\mathbf{Y}},$$

$$\text{i.e., } \hat{\varepsilon}_{ik} = y_{ik} - \hat{y}_{ik}, \quad i = 1, \dots, n, \quad k = 1, \dots, d$$

Hat Matrix

Just like in univariate linear regression we can write the fitted values as

$$\begin{aligned}\hat{\mathbf{Y}} &= \mathbf{X}\hat{\mathbf{B}} \\ &= \mathbf{X}(\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\mathbf{Y} \\ &= \mathbf{H}\mathbf{Y},\end{aligned}$$

where $\mathbf{H} = \mathbf{X}(\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T$ is the **hat matrix**

$\Rightarrow \mathbf{H}$ projects $\mathbf{y}_k$ onto the column space of $\mathbf{X}$ for $k \in \{1, \dots, d\}$

Partitioning the Total Variation

We can partition the total covariation in $\{\mathbf{y}_i\}_{i=1}^n$ (SSCP_{Tot}) as

$$\begin{aligned}\text{SSCP}_{\text{tot}} &= \sum_{i=1}^n (\mathbf{y}_i - \bar{\mathbf{y}})^T (\mathbf{y}_i - \bar{\mathbf{y}}) \\&= \sum_{i=1}^n (\mathbf{y}_i - \hat{\mathbf{y}}_i + \hat{\mathbf{y}}_i - \bar{\mathbf{y}})(\mathbf{y}_i - \hat{\mathbf{y}}_i + \hat{\mathbf{y}}_i - \bar{\mathbf{y}})^T \\&= \underbrace{\sum_{i=1}^n (\hat{\mathbf{y}}_i - \bar{\mathbf{y}})(\hat{\mathbf{y}}_i - \bar{\mathbf{y}})^T}_{\text{SSCP}_{\text{Reg}}} + \underbrace{\sum_{i=1}^n (\mathbf{y}_i - \hat{\mathbf{y}}_i)(\mathbf{y}_i - \hat{\mathbf{y}}_i)^T}_{\text{SSCP}_{\text{Err}}} \\&\quad + \underbrace{2 \sum_{i=1}^n (\hat{\mathbf{y}}_i - \bar{\mathbf{y}})(\mathbf{y}_i - \hat{\mathbf{y}}_i)}_{=0} = \text{SSCP}_{\text{Reg}} + \text{SSCP}_{\text{Err}}\end{aligned}$$

The corresponding **degrees of freedom** are $d(n-1)$ for SSCP_{Tot} ; dp for SSCP_{Reg} ; and $d(n-p-1)$ for SSCP_{Err}

Estimated Error Covariance

The estimated error covariance matrix is

$$\begin{aligned}\hat{\Sigma} &= \frac{\sum_{i=1}^n (\mathbf{y}_i - \hat{\mathbf{y}}_i)(\mathbf{y}_i - \hat{\mathbf{y}}_i)^T}{n - p - 1} \\ &= \frac{\text{SSCP}_{Err}}{n - p - 1}\end{aligned}$$

- ▶ $\hat{\Sigma}$ is an unbiased estimate of Σ
- ▶ The estimate $\hat{\Sigma}$ is the mean SSCP_{Err}

Sampling Distributions of $\hat{\mathbf{B}}$, $\hat{\mathbf{Y}}$, and $\hat{\mathbf{E}}$

We would need to figure out the **sampling distributions** of estimator and predictor in order to drawn inference

Given the model assumptions, we have

$$\text{vec}(\hat{\mathbf{B}}) \sim \text{N}(\text{vec}(\mathbf{B}), \boldsymbol{\Sigma} \otimes (\mathbf{X}^T \mathbf{X})^{-1})$$

$$\text{vec}(\hat{\mathbf{Y}}) \sim \text{N}(\text{vec}(\mathbf{X}\mathbf{B}), \boldsymbol{\Sigma} \otimes \mathbf{H})$$

$$\text{vec}(\hat{\mathbf{E}}) \sim \text{N}(\mathbf{0}, \boldsymbol{\Sigma} \otimes (\mathbf{I} - \mathbf{H})),$$

where $\text{vec}(\cdot)$ is the vectorization operator and $\otimes$ is the Kronecker product

What Question Does Multivariate Inference Answer?

Suppose predictor X_j is cyl.

Separate regressions ask:

Does cyl affect mpg?

Does cyl affect disp?

Does cyl affect hp?

Does cyl affect wt?

Multivariate regression can ask one joint question:

$$H_0 : \beta_{j1} = \beta_{j2} = \cdots = \beta_{jd} = 0.$$

Does this predictor contribute to the response vector as a whole?

Inference about Multiple $\hat{\beta}_{jk}$

Assume that $q < p$ and want to test if a reduced model is sufficient:

$$H_0 : \mathbf{B}_2 = \mathbf{0}_{p-q} \times d, \quad \text{versus} \quad H_a : \mathbf{B}_2 \neq \mathbf{0}_{p-q} \times d,$$

where

$$\mathbf{B} = \begin{bmatrix} \mathbf{B}_1 \\ \mathbf{B}_2 \end{bmatrix}$$

is the partitioned of the coefficient vector

We can compare the SSCP_{Err} for the **full model**:

$$y_{ik} = \beta_{0k} + \sum_{j=1}^p \beta_{jk} x_{ij} + \varepsilon_{ik}, \quad k = 1, \dots, d$$

and the **reduced model**:

$$y_{ik} = \beta_{0k} + \sum_{j=1}^q \beta_{jk} x_{ij} + \varepsilon_{ik}, \quad k = 1, \dots, d$$

Some Test Statistics

Let $\tilde{\mathbf{E}} = n\tilde{\mathbf{\Sigma}}$ denote the SSCP_{Err} matrix from the **full model**, and let $\tilde{\mathbf{H}} = n(\tilde{\mathbf{\Sigma}}_1 - \tilde{\mathbf{\Sigma}})$ denote the hypothesis SSCP_{Err} matrix

Some test statistics for

$$H_0 : \mathbf{B}_2 = \mathbf{0}_{p-q} \times d, \quad \text{versus} \quad H_a : \mathbf{B}_2 \neq \mathbf{0}_{p-q} \times d :$$

► Wilks's Lambda

$$\Lambda^* = \frac{|\tilde{\mathbf{E}}|}{|\tilde{\mathbf{H}} + \tilde{\mathbf{E}}|}$$

Reject H_0 if Λ^* is “small”

► Hotelling-Lawley Trace

$$T_0^2 = \text{tr}(\tilde{\mathbf{H}}\tilde{\mathbf{E}}^{-1})$$

Reject H_0 if T_0^2 is “large”

► Pillai's Trace

$$V = \text{tr}(\tilde{\mathbf{H}}(\tilde{\mathbf{H}} + \tilde{\mathbf{E}})^{-1})$$

Interval Estimation

We would like to estimate the **expected value of the response** for a given predictor $\mathbf{x}_h = (1, x_{h1}, \dots, x_{hp})$.

Note that we have

$$\hat{\mathbf{y}}_h \sim N(\mathbf{B}^T \mathbf{x}_h, \mathbf{x}_h^T (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{x}_h \Sigma)$$

We can exploit the duality between interval estimation and hypothesis testing. That is, we can test

$$H_0 : \mathbb{E}(\mathbf{y}_h) = \mathbf{y}_h^* \text{ versus } H_a : \mathbb{E}(\mathbf{y}_h) \neq \mathbf{y}_h^*$$

The $100(1 - \alpha)\%$ confidence region is the collection of $\mathbf{y}_h^*$ values that fail to reject H_0 at α level

Interval Estimation (Cont'd)

Test statistics:

$$T^2 = \left(\frac{\hat{\mathbf{B}}^T \mathbf{x}_h - \mathbf{B}^T \mathbf{x}_h}{\sqrt{\mathbf{x}_h^T (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{x}_h}} \right)^T \hat{\Sigma}^{-1} \left(\frac{\hat{\mathbf{B}}^T \mathbf{x}_h - \mathbf{B}^T \mathbf{x}_h}{\sqrt{\mathbf{x}_h^T (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{x}_h}} \right)$$
$$\underset{H_0}{\sim} \frac{d(n-p-1)}{n-p-d} F_{d, n-p-d}$$

Therefore, the $100(1 - \alpha)\%$ simultaneous **confidence interval** for y_{hk} is

$$\hat{y}_{hk} \pm \sqrt{\frac{d(n-p-1)}{n-p-d} F_{d, n-p-d}} \sqrt{\mathbf{x}_h^T (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{x}_h} \hat{\sigma}_{kk},$$

$$k \in \{1, \dots, d\}$$

Predicting New Observations

Here we want to predict the **observed value of response** for a given predictor

- ▶ **Note:** interested in actual $\hat{\mathbf{y}}_h$ instead of $\mathbb{E}(\hat{\mathbf{y}}_h)$
- ▶ Given $\mathbf{x}_h = (1, x_{h1}, \dots, x_{hp})$, the fitted value is still $\hat{\mathbf{y}}_h = \hat{\mathbf{B}}^T \mathbf{x}_h$

We can exploit the duality between interval estimation and hypothesis testing. That is, we can test

$$H_0 : \mathbf{y}_h = \mathbf{y}_h^* \text{ versus } H_a : \mathbf{y}_h \neq \mathbf{y}_h^*$$

The $100(1 - \alpha)\%$ prediction interval is the collection of $\mathbf{y}_h^*$ values that fail to reject H_0 at α level

Predicting New Observations (Cont'd)

Test statistics:

$$T^2 = \left(\frac{\hat{\mathbf{B}}^T \mathbf{x}_h - \mathbf{B}^T \mathbf{x}_h}{\sqrt{1 + \mathbf{x}_h^T (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{x}_h}} \right)^T \hat{\Sigma}^{-1} \left(\frac{\hat{\mathbf{B}}^T \mathbf{x}_h - \mathbf{B}^T \mathbf{x}_h}{\sqrt{1 + \mathbf{x}_h^T (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{x}_h}} \right)$$
$$\underset{H_0}{\sim} \frac{d(n-p-1)}{n-p-d} F_{d, n-p-d}$$

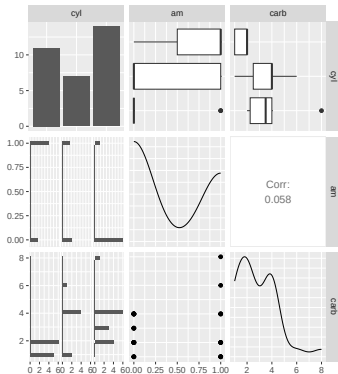
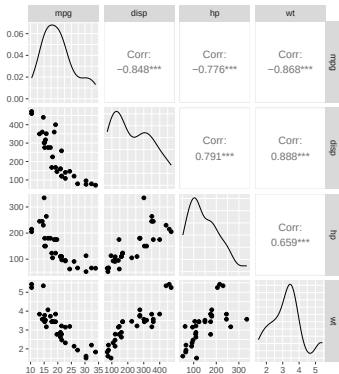
Therefore, the $100(1 - \alpha)\%$ simultaneous **prediction interval** for y_{hk} is

$$\hat{y}_{hk} \pm \sqrt{\frac{d(n-p-1)}{n-p-d} F_{d, n-p-d} \sqrt{(1 + \mathbf{x}_h^T (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{x}_h) \hat{\sigma}_{kk}}},$$

$$k \in \{1, \dots, d\}$$

Motor Trend Car Road Tests Data Analysis

Study the linear relationship between mpg, disp, hp, wt (responses) and cyl, am, carb (predictors) in the mtcars dataset



Multivariate Regression Fit

- ▶ Model: `lm(Y ~ cyl + am + carb, data = mtcars)`
- ▶ Key findings:
 - mpg: decreases with more cylinders, increases with manual transmission
 - disp: strongly influenced by cyl
 - hp: influenced by cyl and carb
 - wt: influenced by cyl, am, carb
- ▶ **Note:** Multivariate regression produces the *same point estimates* as running separate regressions for each response

SSCP & Error Covariance

SSCP decomposition:

$$SSCP_{Tot} = SSCP_{Reg} + SSCP_{Err}$$

- Estimated error covariance matrix ($\hat{\Sigma}$)

	mpg	disp	hp	wt
mpg	7.8680094	-53.27166	-19.7015979	-0.6575443
disp	-53.2716607	2504.87095	425.1328988	18.1065416
hp	-19.7015979	425.13290	577.2703337	0.4662491
wt	-0.6575443	18.10654	0.4662491	0.2573503
	mpg	disp	hp	wt
mpg	1.00000000	-0.3794645	-0.29233405	-0.46209388
disp	-0.3794645	1.00000000	0.35354314	0.71314929
hp	-0.2923340	0.3535431	1.00000000	0.03825304
wt	-0.4620939	0.7131493	0.03825304	1.00000000

⇒ Captures residual dependencies among responses

Do We Need "cyl"?

```
> mv1m0 <- lm(Y ~ am + carb, data = mtcars)
> anova(mv1m, mv1m0, test = "Wilks")
```

Analysis of Variance Table

Model 1: $Y \sim \text{cyl} + \text{am} + \text{carb}$

Model 2: $Y \sim \text{am} + \text{carb}$

	Res.Df	Df	Gen.var.	Wilks	approx F	num Df
1	27		29.862			
2	29	2	43.692	0.16395	8.8181	8

	den	Df	Pr(>F)
1			
2	48		2.525e-07 ***

Interpretation: cylinder count explains variation across responses

Confidence and Prediction Intervals

Default `predict()` lacks multivariate CI/PI, so we use an R function from Prof. Helwig at the University of Minnesota

```
> newdata <- data.frame(cyl = factor(6, levels = c(4, 6, 8)),
+                       am = 1, carb = 4)
> # confidence interval
> pred.mlm(mvmlm, newdata)

      mpg      disp      hp      wt
fit 21.51824 159.2707 136.98500 2.631108
lwr 16.65593  72.5141  95.33649 1.751736
upr 26.38055 246.0273 178.63351 3.510479
> # prediction interval
> pred.mlm(mvmlm, newdata, interval = "prediction")

      mpg      disp      hp      wt
fit 21.518240 159.27070 136.98500 2.6311076
lwr  9.680053 -51.95435  35.58397 0.4901152
upr 33.356426 370.49576 238.38603 4.7720999
```

Take-Home Message

1. Multiple responses are modeled together:

$$Y = XB + E.$$

2. OLS coefficients are the **same** as fitting separate regressions.
3. The multivariate model additionally captures **residual dependence** through Σ .
4. That dependence enables **joint inference** about predictors across several responses.

Fit separately; understand and infer jointly.